

WORDGRAPH · REQUIREMENTS & STORIES

Functional Requirements & User Stories

Visual Vocabulary Explorer

This document defines what WordGraph must do, for whom, and how success is measured. It covers all shipped features and proposed work organised by functional area and user persona.

DOCUMENT TYPE

VERSION

STATUS

FR + User Stories

1.0

Living document

Scope & Conventions

Requirements use MoSCoW priority labels. User stories follow the format *"As a [persona], I want [goal] so that [reason]"* with explicit acceptance criteria. Each story maps to one or more functional requirements.

Label	Meaning
MUST	Non-negotiable — the product does not ship without this
SHOULD	High-value — included unless time or complexity prevents it
COULD	Nice-to-have — included if capacity allows

FUNCTIONAL AREAS

- Search & Discovery
- Graph
- Library
- Data Ownership
- UX & Accessibility

USER PERSONAS



The Writer
Novelist, copywriter,
journalist



The Learner
ESL student, self-
taught reader



The Thinker
Researcher,
knowledge worker



The Browser
Curious reader, in-
the-moment

Functional Requirements

Search & Discovery

FR-SD-01

Search any English word

The system shall look up any English word entered by the user and return its definition, pronunciation, part of speech, synonyms, antonyms, and related words. Lookup shall use the Free Dictionary API as the primary source and Datamuse API as a fallback for words not found in the primary source.

MUST

FR-SD-02

Cache lookups for offline use

Successful dictionary API responses shall be stored in IndexedDB with a 7-day TTL. On subsequent visits to the same word, the cached result shall be returned instantly without a network request. Confirmed "not found" responses shall also be cached to avoid repeated failed fetches.

MUST

FR-SD-03

Show recent words on the home screen

The home screen shall display the last N words the user explored in the current session (stored in sessionStorage). Clicking a recent word shall navigate directly to its explore page.

SHOULD

FR-SD-04

Autocomplete suggestions while typing

The search input shall display a dropdown of word suggestions as the user types (≥ 2 characters). Suggestions shall be drawn in priority order from: (1) the static seed word database, (2) IndexedDB-cached lookups, (3) the Datamuse suggestion endpoint. Selecting a suggestion shall navigate immediately to that word's explore page.

SHOULD

FR-SD-05

Global search from every page

A search input shall be accessible from all pages (home, explore, library, settings) so that users never need to navigate back to the home screen to look up a new word.

SHOULD

FR-SD-06

Graceful error and not-found states

If a word cannot be found in any source, the system shall display a clear "word not found" message with a suggestion to check spelling. If a network error occurs, the system shall display a distinct error state and offer a retry action. The system shall never crash or show a blank page.

MUST

Graph

FR-GR-01

Render a typographic word graph

The explore page shall render an interactive SVG graph with the searched word at the centre and its related words (synonyms, antonyms, related) as satellite text nodes. Nodes shall be typographic labels — not geometric shapes or coloured bubbles. Edges shall be rendered as subtle lines colour-coded by relationship type.

MUST

FR-GR-02

Colour-code relationship types

Three relationship types shall each have a distinct, muted colour: synonyms (green), antonyms (red), related words (blue). A legend showing these three types shall be permanently visible in the top-left corner of the graph SVG, unaffected by pan.

MUST

FR-GR-03

Click a node to explore that word

Clicking any satellite node shall navigate to that word's explore page, making it the new centre of the graph. The URL shall update to reflect the new word. Keyboard users shall be able to activate nodes via Enter or Space.

MUST

FR-GR-04

Exploration breadcrumb trail

A breadcrumb trail shall display the session path of explored words (e.g. Pragmatic → Realistic → Honest). Clicking a breadcrumb item shall navigate back to that word. The trail shall persist for the duration of the browser session.

SHOULD

FR-GR-05

Pan the graph

Users shall be able to drag/pan the graph when it contains many nodes that extend beyond the visible area. A "Reset view" control shall appear when the graph has been panned and return it to the default position on click.

SHOULD

FR-GR-06

Filter multi-word graph nodes

The system shall exclude any synonym, antonym, or related word that contains a space (i.e. is a multi-word phrase) from the graph nodes, since clicking such a node produces confusing results. Single-word entries only shall appear as clickable nodes.

SHOULD

FR-GR-07

Animate graph transitions

Nodes and edges shall animate in when a new word loads. Animations shall be disabled automatically when the user's operating system has reduced-motion enabled (prefers-reduced-motion: reduce).

COULD

Word Information Panel

FR-WI-01

Display full word information

Alongside the graph, the system shall display: word heading, IPA pronunciation, part of speech, commonness badge (common / uncommon), definition, up to two usage examples, a memory trick (where available), and usage notes (where available).

MUST

FR-WI-02

Audio pronunciation playback

Where the Free Dictionary API provides an audio URL for a word's pronunciation, the system shall display a play button. Clicking it shall play the pronunciation audio inline without navigating away.

SHOULD

FR-LI-01

Save and unsave words

From any word's explore page, the user shall be able to save the word to their personal library with a single action. The button label shall toggle to reflect saved state. Unsaving shall remove the word and all associated user data (tags, notes, collection memberships) after confirmation.

MUST

FR-LI-02

Personal notes on saved words

Each saved word shall have a free-text notes field editable from the library view. Changes shall be auto-saved with a short debounce (≤ 700 ms) — no explicit save button required. Notes shall persist across sessions via IndexedDB.

MUST

FR-LI-03

Tags on saved words

Users shall be able to add multiple free-text tags to each saved word. Tags shall be stored normalised (lowercase, trimmed, deduplicated). Tags shall be removable individually. The library shall be filterable by tag via a sidebar.

MUST

FR-LI-04

Collections (named groups of words)

Users shall be able to create, rename, and delete named collections. Saved words shall be assignable to one or more collections. Collections shall appear in the library sidebar for filtering. Deleting a collection shall not delete the words within it.

MUST

FR-LI-05

Library graph of saved words

The library shall include a graph view showing all saved words as a connected network, visualising the relationships between them using the same typographic style as the explore graph.

SHOULD

FR-LI-06

All data persists across sessions

All saved words, tags, notes, collections, and collection memberships shall persist across browser sessions and page refreshes via IndexedDB. No data shall be lost on reload or browser restart (subject to user not clearing site data).

MUST

Data Ownership

FR-DO-01

Export library as JSON

Users shall be able to download a complete JSON snapshot of their library from the Settings page. The snapshot shall include all saved words, their tags, notes, collections, and word-to-collection memberships. The file shall be structured, versioned, and machine-readable.

MUST

FR-DO-02

Export library as Markdown and CSV

Users shall be able to download their library as a human-readable Markdown file and as a CSV spreadsheet, in addition to JSON.

SHOULD

FR-DO-03

Import a JSON snapshot

Users shall be able to upload a previously exported JSON snapshot to restore or merge their library. Import shall use merge semantics: existing words are preserved (tags are merged, notes are not overwritten). No data shall be deleted during import. Re-importing the same file shall be idempotent.

MUST

FR-UX-01

Light and dark mode

The application shall support light and dark colour themes. The default shall follow the user's OS preference. A manual toggle shall allow users to override the default for the current session.

MUST

FR-UX-02

Fully keyboard-navigable

All interactive elements — search, graph nodes, save button, library cards, collection picker, export buttons — shall be reachable and operable by keyboard alone. Focus indicators shall be visible and meet WCAG 2.1 AA contrast requirements.

MUST

FR-UX-03

Screen reader support

All meaningful content and interactive elements shall have appropriate ARIA roles, labels, and live regions. Graph nodes and edges shall have descriptive aria-labels (e.g. "realistic — synonym of pragmatic"). Status changes (saving, loading, error) shall be announced to assistive technology.

MUST

FR-UX-04

Responsive layout

The application shall be usable on desktop (≥ 1024 px) and mobile (≥ 360 px) viewports. On desktop, the graph and word-info panel shall be displayed side-by-side. On mobile, they shall stack vertically.

MUST

FR-UX-05

Zero telemetry / no account required

The system shall make no network requests except dictionary API lookups. No analytics, crash reporting, or tracking of any kind. No user account or login shall be required for any feature.

MUST

User Stories

Stories are grouped by persona. Each includes the goal, motivation, and explicit acceptance criteria that define done.

The Writer

US-W-01 Find the exact word to avoid repetition

As a writer,

I want to search a word I've overused and see its synonyms arranged visually, so that I can pick the most precise alternative without breaking my writing flow.

ACCEPTANCE CRITERIA

- ✓ Searching a word loads its graph within 1.5 seconds on a fresh lookup
- ✓ Synonyms are displayed as clickable nodes coloured distinctly from antonyms and related words
- ✓ Clicking a synonym node loads that word's graph so I can compare options
- ✓ The definition and part of speech are visible without scrolling on desktop

US-W-02 Build a personal reference of favourite words

As a writer,

I want to save words I discover during a session and group them by project or theme, so that I can return to my personal word palette during editing without re-searching.

ACCEPTANCE CRITERIA

- ✓ A "Save" button on the explore page adds the word to the library in one click
- ✓ I can create a named collection (e.g. "Novel #2") and assign saved words to it
- ✓ The library persists across browser sessions — closing and reopening the tab retains all saved words
- ✓ I can filter the library to see only words in a specific collection

As a writer,

I want to export my library as a Markdown or CSV file,
so that I can paste words and notes into my manuscript document or writing app.

ACCEPTANCE CRITERIA

- ✓ The Settings page offers Markdown and CSV download options
- ✓ The Markdown export includes each word as a heading with its definition and my personal notes
- ✓ The CSV includes word, definition, tags, and notes as columns
- ✓ Downloads trigger immediately with a sensibly named file

□ The Learner

US-L-01 Understand a new word deeply, not just its definition

As a language learner,

I want to see a new word's pronunciation, definition, examples, and related words all in one place,

so that I can build a complete mental model of the word rather than memorising an isolated definition.

ACCEPTANCE CRITERIA

- ✓ The word panel shows IPA pronunciation, part of speech, definition, and at least one usage example
- ✓ A play button is available when audio pronunciation data exists
- ✓ The graph shows related words I can explore to understand the word's context
- ✓ A memory trick is shown for seed words to aid retention

US-L-02 Add a personal note to remember how to use a word

As a language learner,

I want to write a personal note on a saved word (e.g. a sentence I wrote using it), so that I can recall my own context for the word rather than just the dictionary definition.

ACCEPTANCE CRITERIA

- ✓ Expanding a saved word in the library reveals a notes text area
- ✓ I can type freely and the note saves automatically — no save button needed
- ✓ The note is still there after closing and reopening the browser
- ✓ Notes are included in the JSON export and restored on import

As a language learner,

I want to tag saved words with topic labels (e.g. "academic", "emotions"),
so that I can study related vocabulary groups together.

ACCEPTANCE CRITERIA

- ✓ I can type a tag name and press Enter to add it to a saved word
- ✓ Tags appear as removable pills on the word card
- ✓ Clicking a tag in the library sidebar filters to all words with that tag
- ✓ Tags survive export and import with full fidelity

□ The Thinker

US-T-01 Look up rare or abstract words without hitting a dead end

As a researcher,

I want abstract or philosophical words (like "nothingness" or "serendipity") to return a useful graph,

so that I can explore conceptual vocabulary the same way I explore everyday words.

ACCEPTANCE CRITERIA

- ✓ When the primary dictionary API fails for a word, a second source (Datamuse) is automatically tried
- ✓ The error state ("Something went wrong") is only shown when both sources fail
- ✓ A successful Datamuse result is cached and shown as a normal graph
- ✓ The "not found" state is shown only when both sources confirm the word doesn't exist

US-T-02 Build and maintain a personal vocabulary corpus

As a knowledge worker,

I want to save hundreds of words, organise them into named collections, and export a complete backup,

so that I have a growing, exportable vocabulary reference I own and control.

ACCEPTANCE CRITERIA

- ✓ There is no limit on the number of saved words or collections
- ✓ Exported JSON includes every word, tag, note, collection, and membership
- ✓ Importing the JSON on a fresh browser restores the complete library exactly
- ✓ Re-importing the same file a second time does not duplicate any data

As a researcher,

I want to see a graph of all my saved words and their relationships,
so that I can spot conceptual clusters and gaps in my vocabulary collection.

ACCEPTANCE CRITERIA

- ✓ The library includes a "graph" view showing all saved words as nodes
- ✓ Edges connect words that share synonym/antonym/related relationships
- ✓ The graph uses the same typographic style as the explore graph

□ The Browser

US-B-01 Look up an unfamiliar word without creating an account

As a casual reader,

I want to quickly look up a word I just encountered and see what it means and connects to,

so that I can satisfy my curiosity without signing up for anything or leaving the page I'm reading for long.

ACCEPTANCE CRITERIA

- ✓ No account, login, or email is required to use any part of the app
- ✓ Typing a word and pressing Enter navigates to the explore page in under 2 seconds
- ✓ The word's definition is visible immediately — no interaction required to see it
- ✓ The graph loads alongside the definition without a separate action

US-B-02 Find the word I'm thinking of even if I'm not sure of the spelling

As a casual reader,

I want to see word suggestions appear as I type so I can select the right word, so that I don't hit a "not found" error because of a spelling mistake.

ACCEPTANCE CRITERIA

- ✓ A dropdown of suggestions appears after I type 2 or more characters
- ✓ Suggestions include words I've previously looked up as well as new suggestions
- ✓ I can select a suggestion with mouse, touch, or keyboard arrow keys + Enter
- ✓ The dropdown closes if I click outside it or press Escape

As a curious reader,

I want to click through connected words and always know how I got here,
so that I can trace my exploration path and jump back to an earlier word if I get
sidetracked.

ACCEPTANCE CRITERIA

- ✓ Each word I explore is added to a breadcrumb trail at the top of the page
- ✓ Clicking any breadcrumb item navigates back to that word's graph
- ✓ The breadcrumb persists for the duration of my browser session
- ✓ The URL updates with each new word so I can share or bookmark a specific word

Requirement ↔ Story Traceability

Each functional requirement maps to at least one user story. Use this table to confirm coverage and to scope regression tests.

Requirement	Description	Covered by
FR-SD-01	Search any English word	US-W-01, US-L-01, US-B-01
FR-SD-02	Cache lookups offline	US-T-01, US-B-01
FR-SD-03	Recent words on home	US-B-03
FR-SD-04	Autocomplete suggestions	US-B-02
FR-SD-05	Global search from every page	US-W-01, US-B-02
FR-SD-06	Graceful error / not-found states	US-T-01, US-B-01
FR-GR-01	Typographic word graph	US-W-01, US-L-01, US-B-01
FR-GR-02	Colour-coded relationships + legend	US-W-01, US-L-01
FR-GR-03	Click node to explore	US-W-01, US-B-03
FR-GR-04	Exploration breadcrumb	US-B-03
FR-GR-05	Pan the graph	US-W-01, US-T-03
FR-GR-06	Filter multi-word nodes	US-W-01, US-L-01
FR-GR-07	Animate transitions	US-B-01
FR-WI-01	Full word information panel	US-L-01, US-B-01
FR-WI-02	Audio pronunciation	US-L-01
FR-LI-01	Save / unsave words	US-W-02, US-T-02
FR-LI-02	Personal notes	US-L-02, US-T-02
FR-LI-03	Tags	US-L-03, US-T-02
FR-LI-04	Collections	US-W-02, US-T-02

Requirement	Description	Covered by
FR-LI-05	Library graph	US-T-03
FR-LI-06	Data persists across sessions	US-W-02, US-L-02, US-T-02
FR-DO-01	Export JSON	US-T-02, US-W-03
FR-DO-02	Export Markdown + CSV	US-W-03
FR-DO-03	Import JSON (merge, idempotent)	US-T-02
FR-UX-01	Light / dark mode	All personas
FR-UX-02	Keyboard navigation	All personas
FR-UX-03	Screen reader support	All personas
FR-UX-04	Responsive layout	US-B-01, US-B-03
FR-UX-05	Zero telemetry, no account	US-B-01, All personas